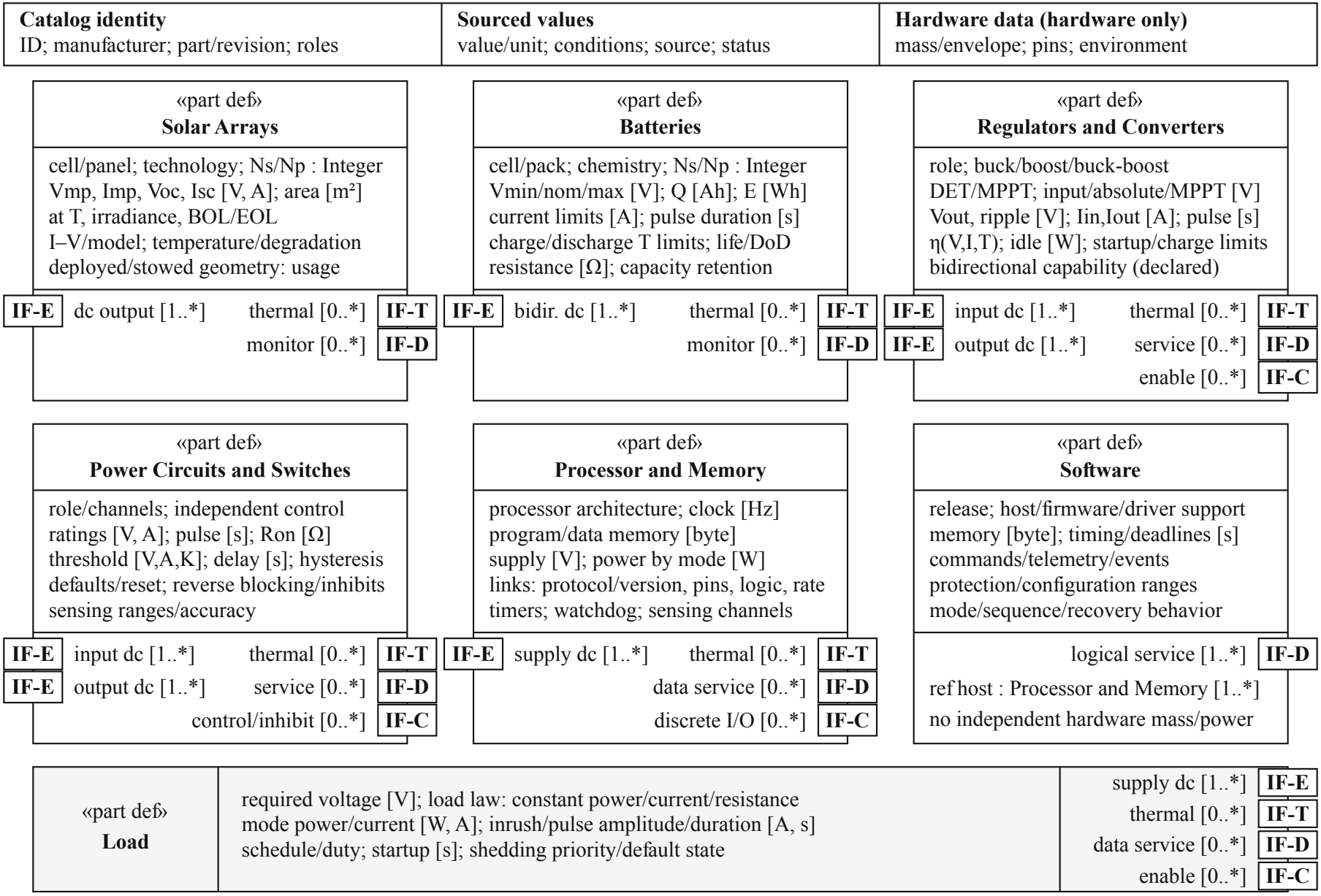

Catalog identity
ID; manufacturer; part/revision; roles
Sourced values
value/unit; conditions; source; status
Hardware data (hardware only)
mass/envelope; pins; environment
«part def»
Solar Arrays
cell/panel; technology; Ns/Np : Integer
Vmp, Imp, Voc, Isc [V, A]; area [m²]
at T, irradiance, BOL/EOL
I–V/model; temperature/degradation
deployed/stowed geometry: usage
IF-E
dc output [1..*]
thermal [0..*]
IF-T
monitor [0..*]
IF-D
«part def»
Batteries
cell/pack; chemistry; Ns/Np : Integer
Vmin/nom/max [V]; Q [Ah]; E [Wh]
current limits [A]; pulse duration [s]
charge/discharge T limits; life/DoD
resistance [Ω]; capacity retention
IF-E
bidir. dc [1..*]
thermal [0..*]
IF-T
monitor [0..*]
IF-D
«part def»
Regulators and Converters
role; buck/boost/buck-boost
DET/MPPT; input/absolute/MPPT [V]
Vout, ripple [V]; Iin,Iout [A]; pulse [s]
η(V,I,T); idle [W]; startup/charge limits
bidirectional capability (declared)
IF-E
input dc [1..*]
thermal [0..*]
IF-T
IF-E
output dc [1..*]
service [0..*]
IF-D
enable [0..*]
IF-C
«part def»
Power Circuits and Switches
role/channels; independent control
ratings [V, A]; pulse [s]; Ron [Ω]
threshold [V,A,K]; delay [s]; hysteresis
defaults/reset; reverse blocking/inhibits
sensing ranges/accuracy
IF-E
input dc [1..*]
thermal [0..*]
IF-T
IF-E
output dc [1..*]
service [0..*]
IF-D
control/inhibit [0..*]
IF-C
«part def»
Processor and Memory
processor architecture; clock [Hz]
program/data memory [byte]
supply [V]; power by mode [W]
links: protocol/version, pins, logic, rate
timers; watchdog; sensing channels
IF-E
supply dc [1..*]
thermal [0..*]
IF-T
data service [0..*]
IF-D
discrete I/O [0..*]
IF-C
«part def»
Software
release; host/firmware/driver support
memory [byte]; timing/deadlines [s]
commands/telemetry/events
protection/configuration ranges
mode/sequence/recovery behavior
logical service [1..*]
IF-D
ref host : Processor and Memory [1..*]
no independent hardware mass/power
«part def»
Load
required voltage [V]; load law: constant power/current/resistance
mode power/current [W, A]; inrush/pulse amplitude/duration [A, s]
schedule/duty; startup [s]; shedding priority/default state
supply dc [1..*]
IF-E
thermal [0..*]
IF-T
data service [0..*]
IF-D
enable [0..*]
IF-C